Stochastic regulation of gene expression

Jeff Hasty, Joel Pradines, Milos Dolnik, and J. J. Collins

Citation: AIP Conference Proceedings **502**, 191 (2000); doi: 10.1063/1.1302384
View online: http://dx.doi.org/10.1063/1.1302384
View Table of Contents: http://aip.scitation.org/toc/apc/502/1
Published by the American Institute of Physics



HTML ABSTRACT

Stochastic Regulation of Gene Expression

Jeff Hasty*, Joel Pradines*, Milos Dolnik† and J.J. Collins*

*Center for BioDynamics and Dept. of Biomedical Engineering
Boston University
Boston, MA 02215, USA
†Dept. of Chemistry and Center for Complex Systems
Brandeis University
Waltham, MA 02454, USA

Abstract. We analyze the effect of additive noise on a previously proposed biochemical model describing the genetic regulation of the production of protein. The steady-state mean value of protein concentration is obtained through the equilibrium solution of the Fokker-Planck equation. In this way it is shown how external noise can serve as a type of control on the overall protein concentration. We demonstrate the utility of such control through the construction of a protein switch, whereby protein production is abruptly altered using short pulses in the magnitude of the external noise source.

INTRODUCTION

Regulated gene expression is the process through which cells control fundamental functions, such as the regulation of enzyme activities through their concentration [1], the production of structural proteins and the time sequence of this production during development [2]. A large part of these regulatory processes takes place at the level of gene transcription [3], and it has been shown theoretically and experimentally that external influences from the environment and other cells can affect the underlying reactions in various ways [4].

As experimental techniques are increasingly capable of providing reliable data pertaining to gene regulation, theoretical models are becoming important in the understanding and manipulation of such processes. The most common theoretical approach is to model the interactions of elements in a regulatory network as biochemical reactions. If one ignores internal fluctuations, rate equations emerge as a first approximation to the general Master-Equation approach describing the chemical reactions. These deterministic differential equations describe the evolution of the mean value of some property of the set of reactions, typically the concentrations of the various elements involved. The existence of positive or negative feedback in a

CP502, *Stochastic and Chaotic Dynamics in the Lakes: STOCHAOS,*
edited by D. S. Broomhead, E. A. Luchinskaya, P.V. E. McClintock, and T. Mullin
© 2000 American Institute of Physics 1-56396-915-7/00/$17.00

regulatory network is thought to be common [6], and within the reaction framework feedback leads to nonlinear rate equations.

Noise in the form of random fluctuations arises in these systems in one of two ways. *Internal* noise is inherent in the biochemical reactions. Its magnitude is proportional to the inverse of the system size, and its origin is often thermal. On the other hand, *external* noise originates in the random variation of one or more of the externally set control parameters, such as the rate constants associated with a given set of reactions. If the noise source is small, its effect can often be incorporated *post hoc* into the rate equations. In the case of internal noise, this is done in an attempt to recapture the lost information embodied in the rate equation approximation. But in the case of external noise, one often wishes to introduce some new phenomenon where the details of the effect are not precisely known. In either case, the governing rate equations are augmented with additive or multiplicative stochastic terms. Viewed as a random perturbation to the deterministic picture, these terms can induce various effects, most notably the switching between potential attractors (i.e. fixed points, periodicity, chaos).

The present work elucidates the behavior of a gene regulatory system in the presence of an external noise source. We do so by allowing rapid fluctuations in one of the control parameters of a simple model describing gene expression.

BACKGROUND

We consider a previously proposed model of biochemical reactions describing the regulation of the production of a protein [7,8]. Let X denote the protein and D a specific binding site on a DNA promoter. The production of X takes place when the dimer X_2 forms a complex with the DNA at site D. Symbolically, we have

$$\begin{aligned} 2X &\underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} X_2 \\ D + X_2 &\underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} DX_2 \\ DX_2 + P &\overset{k_3}{\rightarrow} DX_2 + nX \\ X &\overset{k_4}{\rightarrow} A \end{aligned} \tag{1}$$

The first reaction represents the dimerization of X, and the second the formation of a DNA-promoter complex. In the presence of RNA polymerase P and the reactants required for transcription, this complex leads to the effective transcription of the gene coding for X. The produced mRNA is then translated into protein X, where n is the number of proteins per transcript. Both transcription and translation are represented by the third reaction. Lastly, X degrades via reaction four.

If we consider an *in vitro* system with purified individual elements, then we can take the concentrations of the above elements as our dynamical variables. Further, the first two reactions in Eqs. (1) are orders of magnitude faster than transcription and degradation. This enables a reduction from four rate equations to one

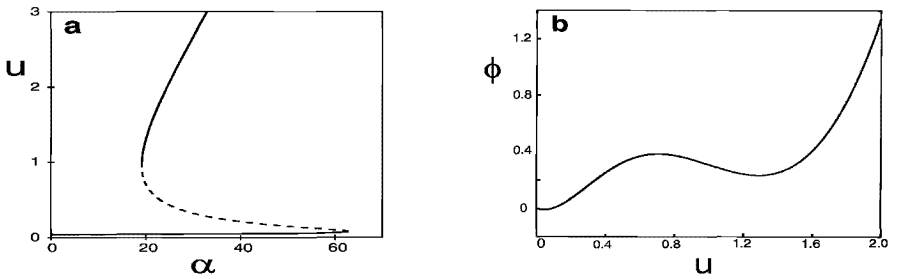


FIGURE 1. (a) Bifurcation plot for the variable u, where u is the concentration of protein X, versus model parameters α. The dotted line correspond to the set of unstable fixed points. (b) The energy landscape corresponding to parameter values $\alpha = 20$, $\beta = 1$, $\gamma = 10$, and $\delta = 0.4$. Stable equilibrium concentration values of Eq. (2) correspond to the valleys at $u = 0.04$ and $u = 1.29$, with an unstable value at $u = 0.71$.

describing the dynamics of the concentration of the protein monomer X

$$\dot{u} = \frac{\alpha u^2}{1 + \beta u^2} - \gamma u + \delta \tag{2}$$

where

$$\alpha = nk_3 p_0 K_1 K_2 d_T, \quad \beta = K_1 K_2, \quad \gamma = k_4, \quad \delta = r_{1bas}$$

The parameter d_T is the total concentration of DNA binding sites, and the equilibrium constants describing the fast reactions are $K_1 = k_1/k_{-1}$ and $K_2 = k_2/k_{-2}$. The additive term $\delta = r_{1bas}$ is the basal rate of production of protein, i.e., the low baseline expression rate in the absence of a transcription factor. The parameters in Eq. (2) determine the steady-state concentration, and for this equation there are two interesting situations. For one set of parameter values, there is monostability, whereby all initial concentrations evolve to the same fixed-point value. For another set, there are three fixed points, and the initial concentration determines which steady state is selected. In Fig. 1a, we demonstrate this behavior by plotting the steady-state concentration u_{ss} versus the parameter α. This bifurcation plot captures the essential dynamics of Eq. (2). It can be seen that for certain values of α there is one unique steady-state value of u, and that for other values there are three. Additionally, in the multiple fixed-point regime, stability analysis indicates that the middle fixed point is unstable, so that all initial values less than the middle fixed point evolve to the lower fixed point, while those greater than the middle fixed point evolve to the upper one. This bistability arises as a consequence of both the dimerization of the protein and the competition between the production via transcription and the degradation in the last reaction of Eqs. (1).

ADDITIVE NOISE

We now focus on parameter values leading to bistability, and consider how an external noise source could affect the production of protein. Physically, we take the dynamical variable u described above to represent the protein concentration within a colony of cells, and consider the noise to act on many copies of this colony. In the absence of noise, each colony will evolve identically to one of the two fixed points, as discussed above. The presence of a noise source will at times modify this simple behavior, whereby colony-to-colony fluctuations can induce novel behavior.

We examine the effect of a noise source which alters the "background" protein production. We posit that such an effect will be small and can be treated as a random perturbation to our existing treatment; we envision that events induced will be interactions between the external noise source and the basal production rate, and that this will translate to a rapidly varying background protein production embodied in the external parameter δ. In order to introduce this effect, we generalize the aforementioned model such that random fluctuations enter Eq. (2) linearly,

$$\dot{u} = f(u) + \xi(t) \tag{3}$$

where $f(u)$ is the right-hand side of Eq. (2), and $\xi(t)$ is a rapidly fluctuating random term with zero mean ($< \xi(t) >= 0$). In order to encapsulate the rapid random fluctuations, we make the standard requirement that the autocorrelation be "δ-correlated", i.e., the statistics of $\xi(t)$ are such that $< \xi(t)\xi(t') >= D\delta(t-t')$, with D proportional to the strength of the perturbation.

Eq. (3) can be rewritten as:

$$\dot{u} = -\frac{\partial \phi(u)}{\partial u} + \xi(t) \tag{4}$$

where the potential $\phi(u)$ is introduced:

$$\phi(u) = -\int f(u)du = -\frac{\alpha + \beta\delta}{\beta}u + \frac{\gamma}{2}u^2 + \frac{\alpha}{\beta^{3/2}}\,\mathrm{ArcTan}(\sqrt{\beta}u) \tag{5}$$

$\phi(u)$ can be viewed as an "energy landscape", whereby u is considered the position of a particle moving in the landscape. One such landscape is plotted in Fig. 1b Note that the stable fixed points correspond to the minima of the potential ϕ in Fig. 1b, and the effect of the additive noise term is to cause random kicks to the particle (system state point) lying in one of these minima. On occasion, a sequence of kicks may enable the particle to escape a local minimum and reside in a new valley.

In order to analyze Eq. (4), one typically introduces the probability distribution $P(u,t)$, which is effectively the probability of finding the system in a state with concentration u at time t. Given Eq. (4), a Fokker-Planck (FP) equation for $P(u,t)$ can be constructed [11]. The steady-state solution for this equation is given by

$$P_s(u) = Ae^{-\frac{2}{D}\Phi(u)} \tag{6}$$

where A is a normalization constant determined by requiring the integral of $P_s(u)$ over all u be unity.

Using the steady-state distribution, the steady-state mean (ssm) concentration $< u >_{ss}$ is given by

$$< u >_{ss} = \int_0^\infty u A e^{-\frac{2}{D}\phi(u)} du \tag{7}$$

In Fig. 2a, we plot the steady-state mean concentration as a function of D, obtained by numerically integrating Eq. (7). It can be seen that the ssm concentration increases with D, corresponding to the increasing likelihood of populating the upper state in Fig. 1b. Figure 2a indicates that the external noise can be used to control

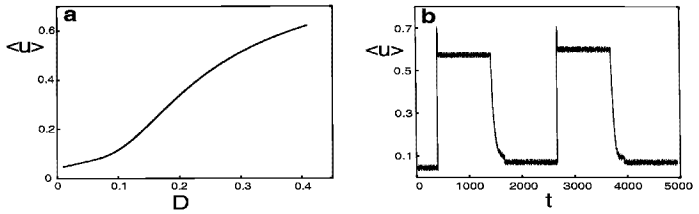


FIGURE 2. Additive noise results. (a) The steady-state equilibrium protein concentration plotted versus noise strength. The concentration increases as the noise causes the upper state of Fig. 1b to become increasingly populated. (b) Simulation of Eq. (3) demonstrating the utilization of external noise for protein switching (see text for details). The parameter values are those of Fig. 1b and the simulation technique is that of Ref. 15.

the ssm concentration. As a candidate application, consider the following protein switch. Given parameter values leading to the landscape of Fig. 1b, we begin the switch in the "off" position by tuning the noise strength to a very low value. This will cause a high population in the lower state, and a correspondingly low value of the concentration. Then at some time later, consider pulsing the system by increasing the noise to some large value for a short period of time, followed by a decrease back to the original low value. The pulse will cause the upper state to become populated, corresponding to a concentration increase and a flipping of the switch to the "on" position. As the pulse quickly subsides, the upper state remains populated as the noise is not of sufficient strength to drive the system across either barrier (on relevant time scales). To return the switch to the off position, the upper-state population needs to be decreased to a low value. This can be achieved by applying a second noise pulse of intermediate strength. This intermediate value is chosen large enough so as to enhance transitions to the lower state, but small enough as to remain prohibitive to upper-state transitions.

Figure 2b depicts the time evolution of the switching process for noise pulses of strengths $D = 0.4$ and $D = 0.06$. Initially, the concentration begins at a level of $< u >_{ss} = 0.06$, corresponding to a low noise value of $D = 0.01$. At time $t = 480$, a noise burst of strength $D = 0.4$ is used to drive the concentration to a value

of $< u >_{ss} \sim 0.6$. Following this burst, the noise is returned to its original value. At $t = 1480$, a second noise burst of strength $D = 0.06$ is used to return the concentration to its original value.

CONCLUSION

These results suggest that an external noise source could be used as a genetic switch. Physically, the noise might be generated using an external field. Importantly, it has been demonstrated experimentally in numerous contexts that electromagnetic fields can exert biological effects [13]. In addition, recent theoretical [10] and experimental [9] work suggests a possible mechanism whereby an electric field can alter an enzyme-catalyzed reaction. These findings suggest that, although there is global charge neutrality, an external field can interact with local dipoles which arise through transient conformational changes or in membrane transport.

Current gene therapy techniques are limited in that transfected genes are typically either in an "on" or "off" state. However, for the effective treatment of many diseases, the expression of a transfected gene needs to be regulated in some systematic fashion. Thus, the development of externally-controllable noise-based switches for gene expression could have significant clinical implications.

ACKNOWLEDGEMENTS

This work was supported by the Office of Naval Research and the U.S. Department of Energy.

REFERENCES

1. R. Dickson, J. Abelson, W. Barnes, *Science* **187**, 27 (1975).
2. C. H. Yuh, H. Bolouri, E. H. Davidson, *Science* **279**, 1896 (1998).
3. B. Lewin, *Genes VI* (Oxford University Press, Oxford, 1997).
4. P. C. Hanawalt, *Mut. Res.* **400**, 117 (1998).
5. H. H. McAdams and A. Arkin, *Proc. Natl. Acad. Sci.* **94**, 814 (1997).
6. H. H. McAdams and L. Shapiro, *Science* **269**, 650 (1995).
7. P. Smolen, D. A. Baxter, J. H. Byrne, *Am. J. Physiol. - Cell Ph.* **43**, C531 (1998).
8. J. Lewis, M. W. Slack, L. Wolpert, *J. Theor. Biol.* **65**, 579 (1977).
9. T. D. Xie, P. Marszalek, Y. Chen, *Biophys. J.* **67**, 1247 (1994).
10. R. D. Astumian and B. Robertson, *J. Am. Chem. Soc.* **115**, 11063 (1993).
11. Van Kampen, N. G. *Stochastic Processes in Physics and Chemistry* (North-Holland, Amsterdam, 1992).
12. W Horsthemke and R. Lefever, *Noise-Induced Transitions* (Springer-Verlag, Berlin, 1984).
13. C. A. Bassett, *J. Cell. Biochem.* **51**, 387 (1993).
14. P. H. von Hippel, *Science* **281**, 660 (1998).
15. J. Sancho, M. S. Miguel, S. Katz, *Phys. Rev. A* **26**, 1589 (1982).